

FREDERICK DOUGLAS PEARCE

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SENIOR AI ENGINEER, MACHINE LEARNING & INTELLIGENT SYSTEMS DEVELOPMENT

Artificial Intelligence ♦ Agentic AI ♦ LLM Applications ♦ Machine Learning Engineering & MLOps ♦ Data Science ♦ Data Analytics

Senior ML and AI engineer with 15+ years of experience building production AI systems, ML pipelines, and data products spanning LLM applications, agentic AI, MLOps, analytics, and cloud software. PhD from MIT with a record of developing published AI products, production ML systems, and data pipelines from concept through deployment, testing, and ongoing operation. Translates complex organizational needs into practical, cost-effective AI/ML data solutions.

- **AI FLUENCY SUITE:** Created and published AgentFluent and CodeFluent, open-source AI products that evaluate autonomous agents and human AI collaboration through research-driven scoring, diagnostics, and developer coaching.
- **SPORTSWEAR ESG NEWS CLASSIFIER:** Designed a fully autonomous production ML pipeline that monitors ESG news, classifies content through hybrid ML and LLM pipelines, and publishes live sustainability intelligence for 50+ global brands.
- **SEMANTICBITS (CMS QUALITY PAYMENT PROGRAM):** Built production data products, executive dashboards, and public reporting powering Medicare quality payment analytics.

ML / AI: Large Language Models, LLM-as-Judge, Agentic AI, RAG, NLP, Text Classification, Clustering, Anomaly Detection, Feature Engineering, XGBoost, Random Forest, Evidently AI, SHAP, Prompt Engineering, Loop Engineering

Languages & Frameworks: Python (FastAPI, Pydantic, NumPy, Pandas, Scikit-learn, Pytest), SQL, MATLAB, TypeScript (VS Code Extension)

Cloud, DevOps, & MLOps: Google Cloud Run, Docker, GitHub Actions, CI/CD, MLFlow

Data & Storage: PostgreSQL, pgvector, Redshift, Elasticsearch

Tools: Claude Code, Claude Agent SDK, Claude API, Jupyter, Superset, VS Code, uv, git

Agile & Collaboration: Scrum, Kanban, Jira, Confluence, GitHub

PROFESSIONAL EXPERIENCE

FREELANCE; PORTLAND, OR (REMOTE)

June 2022 – Present

Freelance AI Engineer / Machine Learning Engineer / Data Scientist

Design, build, and deploy production software, machine learning, and agent-based AI solutions spanning LLM evaluation, autonomous agents, analytics, data engineering, and intelligent automation using Python, SQL, and modern cloud technologies.

AgentFluent — CLI Tool, PyPI Published | v1.1, 2026

April 2026 – Present

- Develop and publish AgentFluent, a local-first Python CLI available on PyPI that analyzes Claude Code and Claude Agent SDK session traces to diagnose agent failures and recommend configuration improvements.

- Create deterministic evaluation pipelines that correlate prompt design, tool selection, model configuration, retry behavior, token usage, and execution errors with actionable optimization recommendations.
- Produce prioritized reports that evaluate agent effectiveness across three axes: 1) configuration quality (rubric-based scoring against Anthropic's documented best practices), 2) behavioral efficiency (retry rates, mid-flight corrections, file rework density), and 3) operational cost (token spend, model selection trade-offs).
- Enhance quality diagnostics with opt-in git and GitHub enrichment layers that correlate session behavior to downstream CI failure rates, PR review density, and feat-to-fix commit proximity.
- Implement rolling baseline regression detection, structured JSON outputs, CI/CD integration, automated versioning, and quality gates using GitHub Actions, mypy, pytest, ruff, and release-please.
- Design a privacy-first, zero-cost architecture that executes entirely on local infrastructure without external APIs, enabling enterprise-safe agent evaluation, monitoring, and continuous improvement.
- **Tech:** Claude Code, Claude Agent SDK, Python, NLP, Clustering, Pydantic, PyYAML, pytest, mypy, ruff, GitHub Actions, Loop Engineering

CodeFluent — VS Code Extension & Web App | v1.1, 2026

February 2026 – Present

- Develop and publish CodeFluent as a VS Code extension and FastAPI web application that analyzes Claude Code session data to measure human-AI collaboration quality against published Anthropic research benchmarks.
- Design LLM-based behavioral evaluation pipelines that score 11 AI fluency behaviors, 6 coding interaction patterns, project configuration maturity while flagging CLAUDE.md rules that lack programmatic enforcement hooks.
- Implement personalized coaching capabilities that generate optimized prompts, identify missing collaboration behaviors, and correlate fluency scores with token consumption, cache utilization, and development cost.
- Architect cross-platform Python and TypeScript services with persistent caching, structured JSON parsing, GitHub CLI integration, and comprehensive automated testing exceeding 1,650 unit and integration tests.
- Build a production-ready analytics platform that transforms raw Claude Code conversations into actionable engineering insights through interactive visualizations, research-backed scoring models, and repository-specific recommendations.
- **Tech:** Python, FastAPI, VS Code Extension API, Claude API, Chart.js, pytest, TypeScript

Sportswear ESG News Classifier | Production ML Pipeline + Live ESG Dashboard

December 2025 – Present

- Architected and deployed a fully autonomous production ML pipeline that monitors ESG news for 50+ global sportswear brands, classifies articles through a hybrid ML and LLM workflow, and publishes a live Sustainability Scorecard with evidence-backed rankings, sentiment analysis, and searchable dashboards.
- Built a two-stage classification pipeline using a Random Forest (F2: 0.974) pre-filter ahead of LLM-as-Judge ESG classification, reducing inference costs by 10% while preserving high recall to capture all relevant content.
- Created a RAG pipeline with OpenAI embeddings, PostgreSQL, and pgvector to retrieve supporting article passages for every ESG classification, providing traceable evidence instead of opaque model outputs.
- Automated end-to-end operations with Python orchestration for news collection, drift detection, model evaluation, website publishing, and health reporting, replacing 5-10 hours of weekly manual effort at an operating cost of about \$0.15/day.
- Deployed and maintained production infrastructure using FastAPI, Docker, Google Cloud Run, MLflow model registry, Evidently AI, GitHub Actions CI/CD, and a 900+ test suite supporting continuous delivery and monitored model performance.
- **Tech:** Python, Claude API, OpenAI Embeddings, scikit-learn, Jupyter, FastAPI, Docker, Google Cloud Run, PostgreSQL, pgvector, MLflow, Evidently AI, GitHub Actions

E-Commerce Fraud Detection System | Production ML Service (ML Zoomcamp)

October 2025 – Present

- Built a production-grade fraud detection service combining feature engineering, model training, explainability, automated testing, and cloud native REST APIs to evaluate real-time payment transactions with configurable business risk thresholds.
- Developed an XGBoost classifier achieving 87% PR AUC with 70% precision at 83% recall on a 299K transaction dataset with a 44:1 class imbalance, generating 30 domain-specific behavioral and transaction features.
- Implemented five configurable decision threshold strategies and TreeSHAP-based prediction explanations, enabling fraud teams to balance false positives against fraud losses while providing transparent, analyst-ready decision rationale.
- Designed and deployed a FastAPI service with Pydantic validation, multi-stage Docker containers, and Google Cloud Run auto-scaling, sustaining 25 requests/second at <70 ms P99 latency under Locust load testing.
- Validated production reliability through a 425 test suite covering preprocessing, feature engineering, model inference, explainability, API endpoints, and integration workflows with 100% critical path coverage.
- **Tech:** Python, XGBoost, scikit-learn, Jupyter, FastAPI, Docker, Google Cloud Run, SHAP, Pydantic, Pytest, Locust

US Presidential Voting Data API | Data Engineering + Public API

December 2021 – Present

- Built a multi-source ETL pipeline ingesting U.S. Electoral College and popular vote data from the National Archives (web scraping), MIT Election Lab (CC0 dataset), and UCSB American Presidency Project, reconciling sources onto shared candidate and state keys in a PostgreSQL star schema warehouse spanning elections from 1824 to present.
- Architected a database-independent API serving layer using read-only SQLite snapshots, allowing Docker containers to serve requests with PostgreSQL offline and enabling scale-to-zero Cloud Run deployment without runtime database dependencies.
- Deployed a production FastAPI REST API behind Cloudflare with CDN caching, rate limiting, and bot protection; built keyless GitHub Actions CI/CD using Workload Identity Federation and version-based cache invalidation for reliable releases.
- Embedded data-governance controls into the architecture, enforcing source-specific licensing requirements, excluding non-redistributable UCSB data from public endpoints, and embedding provenance metadata in every API response.
- **Tech:** Python, FastAPI, PostgreSQL, SQLite, Docker, Google Cloud Run, Cloudflare, GitHub Actions, BeautifulSoup, pandas, geopandas, uv

PORTLAND STATE UNIVERSITY FOUNDATION; PORTLAND, OR

February 2021 – June 2021

Strategic Data Analyst

Completed end-to-end training on Blackbaud CRM and developed working knowledge of the Foundation's donor data environment, including MS SQL Server, SSRS, and SSIS.

- Modified existing SQL queries and SSRS reports to add filters showing each fundraiser's quarterly/campaign results against established fundraising goals, giving the VP of Development visibility into performance by individual and fundraising unit.
- Validated report outputs and underlying data to ensure accurate, reliable results and delivered the reporting solution ahead of deadline, providing actionable intelligence to prioritize fundraising follow-up and help drive progress toward organizational goals.
- Initiated Python-based RFM donor segmentation to analyze donor recency, frequency, and monetary value, establishing an early framework for more targeted donor analysis.

Data Scientist

Served on a three-person team owning the data mart layer, executive dashboards, and public reporting for CMS's Quality Payment Program, which governed Medicare payment outcomes for roughly 1M clinicians.

- Developed production SQL, Python, and Superset data products supporting executive dashboards, clinician lookup tools, policy simulation, ad hoc analytics, and annual CMS reporting.
- Designed and maintained the program's centralized data mart, serving as the required analytical foundation for downstream applications, reporting, and consumer teams.
- Rebuilt the executive dashboard reporting layer during migration from PostgreSQL to Amazon Redshift, validated production readiness, and reduced average page load times by more than 70%.
- Authored complex SQL queries, metrics, and reporting tables for the 2017 and 2018 QPP Experience Reports, delivering accurate public performance reporting under fixed federal deadlines.
- Implemented automated SQL validation routines, verified production datasets, and performed Git based peer reviews of SQL and Python code before release.
- Partnered with data engineering, web development, QA, UI/UX, and business analysis teams to deliver sprint releases, resolve complex data issues, and present completed solutions to CMS program leadership.
- Selected for the hiring team within eight months; conducted candidate screens and technical interviews, designed a BI assessment, and later mentored an incoming data scientist through transition.
- **Tech:** Python, SQL, PostgreSQL, Redshift, Jupyter, Superset, Git, GitHub

TRIPWIRE; PORTLAND, OR

September 2015 – May 2016

Data and Analytics Specialist/Software Engineer

Developed security analytics, anomaly detection, and visualization solutions using Python and the Elastic Stack as Tripwire's first data science hire, transforming raw security event data into actionable product insights.

- Architected a Python analytics framework with 25+ statistical and signal processing features, including FFT and proprietary block-based methods, to characterize security event behavior.
- Built anomaly detection algorithms that identified abnormal security activity by comparing live event patterns against historical baselines across multiple event categories.
- Designed interactive visualizations with Matplotlib to analyze filtered and aggregated Tripwire Enterprise security data, exposing temporal patterns and anomalous events.
- Created a real-time proof of concept simulation demonstrating continuous anomaly detection on streaming security events, enabling product and engineering teams to evaluate practical security analytics capabilities.
- Leveraged Python, Elasticsearch, Logstash, Kibana, NumPy, and Matplotlib to prototype security analytics solutions that expanded Tripwire Enterprise's investigative, visualization, and anomaly detection capabilities.

EDUCATION & PROFESSIONAL DEVELOPMENT

Doctor of Philosophy (Ph.D.), Geophysics | Massachusetts Institute of Technology

Master's Degree, Geosystems | Massachusetts Institute of Technology

Bachelor's Degree, Geological Engineering | University of Idaho

CERTIFICATIONS:

AI Fluency Framework | Anthropic (2026)

ML Zoomcamp Certificate | DataTalksClub (2026)

Practical Data Science Specialization (AWS SageMaker) | Coursera
Deep Learning Specialization | Coursera / DeepLearning.AI
Machine Learning Specialization | Stanford / Coursera
Scrum Master Specialization | Coursera

TECHNICAL WRITING & VOLUNTEER CONTRIBUTIONS

Counted, Not Assumed | **Blog Series** | frederick-douglas-pearce.github.io/blog **2026 - Present**

- Publishing in-depth data journalism examining two centuries of U.S. presidential election outcomes from primary sources: National Archives, MIT Election Lab, and UCSB American Presidency Project. Supported by an automated cross-repo publishing pipeline with fail-closed OG card validation, content-compare idempotency, and validate-all-then-write atomicity guarantees.

Claude Code Session Data | **Blog Series** | frederick-douglas-pearce.github.io/blog **2026 - Present**

- Published 9 in-depth technical articles documenting Claude Code session architecture, tool-use patterns, subagent traces, cost analysis, and evolving data formats, supported by open source utilities for security validation and change detection.

Machine Learning Zoomcamp — Volunteer AI Engineer | **DataTalksClub** **2025**

- Built a RAG-based GitHub automation assistant for DataTalksClub's open source ML community that classified FAQ submissions, detected duplicates, and automated pull request workflows with 153 tests and GitHub Actions deployment.

Scrum Master (Volunteer) | **AgilePDX** **2024**

- Co-organized AgilePDX's AI & Agility Speaker Series, a new program exploring how AI enhances team performance, streamlines processes, and improves product outcomes. Responsible for speaker identification, prioritization, outreach, and presentation logistics across a four-month run.

Technical Interview Evaluator (Volunteer) | **Emerging Leaders PDX** **2021**

- Conducted mock technical interviews and gave structured feedback for college students from underrepresented groups pursuing tech internships, supporting a Portland-area DEI initiative.